

Github: <https://github.com/movchun567/apz-kravchuk.git> , branch micro_mq

Запускаємо docker compose і перевіряємо чи всі компоненти піднялись:

CONTAINER ID	IMAGE	COMMAND	CREATED	STATUS	PORTS	NAMES
3a91ca7ba74c	apz-kravchuk-facade-service	"unicorn -w 4 -b 0..."	49 seconds ago	Up 47 seconds	0.0.0.0:8000->8000/tcp	apz-kravchuk-facade-service-1
e773d8c56920	apz-kravchuk-counter-service	"unicorn -w 2 -b 0..."	49 seconds ago	Up 48 seconds	0.0.0.0:8002->8002/tcp	apz-kravchuk-counter-service-1
323c9ebd1d91	apz-kravchuk-logging-service-2	"unicorn -w 2 -b 0..."	49 seconds ago	Up 48 seconds	0.0.0.0:8003->8001/tcp	apz-kravchuk-logging-service-2-1
a5c6d6ccf865	apz-kravchuk-logging-service-1	"unicorn -w 2 -b 0..."	49 seconds ago	Up 48 seconds	0.0.0.0:8001->8001/tcp	apz-kravchuk-logging-service-1-1
586410ed3cce	apz-kravchuk-logging-service-3	"unicorn -w 2 -b 0..."	49 seconds ago	Up 48 seconds	0.0.0.0:8004->8001/tcp	apz-kravchuk-logging-service-3-1
092230132fcf	hazelcast/hazelcast:5.3	"hz start"	49 seconds ago	Up 48 seconds	5701/tcp	apz-kravchuk-hazelcast3-1
66e89dd127f0	apz-kravchuk-config-server	"unicorn -w 2 -b 0..."	49 seconds ago	Up 48 seconds	0.0.0.0:8010->8010/tcp	apz-kravchuk-config-server-1
fb0ac24af2b1	hazelcast/hazelcast:5.3	"hz start"	49 seconds ago	Up 48 seconds	5701/tcp	apz-kravchuk-hazelcast2-1
c31933f366d5	postgres:16-alpine	"docker-entrypoint.s..."	49 seconds ago	Up 48 seconds	5432/tcp	apz-kravchuk-postgres-1
4256731b428e	apache/kafka:3.9.0	"/_cacert_entrypoint..."	49 seconds ago	Up 48 seconds	0.0.0.0:9092->9092/tcp	kafka
24cc59066a57	hazelcast/hazelcast:5.3	"hz start"	49 seconds ago	Up 48 seconds	5701/tcp	apz-kravchuk-hazelcast1-1

Перевіримо підключення до config-server:

[HTTP](#) http://localhost:8000/user/user1

GET

▼

http://localhost:8010/services

Docs

Params

Authorization

Headers (9)

Body ●

Scripts

Settings

Query Params

Key	Value
Key	Value

Body

Cookies (1)

Headers (5)

Test Results

{ } JSON ▼

▶ Preview

🖼 Visualize ▼

```
1 {
2   "services": {
3     "counter-service": [
4       "http://counter-service:8002"
5     ],
6     "facade-service": [
7       "http://facade-service:8000"
8     ],
9     "logging-service": [
10      "http://logging-service-1:8001",
11      "http://logging-service-2:8001",
12      "http://logging-service-3:8001"
13    ]
14  },
15  "ttl_sec": 120
16 }
```

Як бачимо, всі компоненти піднялись, тому тепер виконаємо тести.

10 транзакцій та логи сервісів:

The screenshot shows a REST client interface with the following details:

- URL:** `http://localhost:8000/transaction`
- Method:** `POST`
- Body Type:** `raw` (selected), with `JSON` as an alternative.
- Request Body:**

```
1  {"user_id": "user1", "amount": 1, "transaction_id": "msg1", "message": "msg1"}
2
```
- Response Body:**

```
1  {
2    "queued": true,
3    "transaction_id": "msg1"
4  }
```

HTTP http://localhost:8000/user/user1

GET http://localhost:8000/user/user1

Docs Params Authorization Headers (7) Body Scripts Settings

Body Cookies (1) Headers (5) Test Results

{ } JSON Preview Visualize

```
1 {
2   "balance": 10,
3   "transactions": [
4     {
5       "amount": 1,
6       "message": "msg2",
7       "timestamp": 1776631138.7859244,
8       "transaction_id": "msg2",
9       "user_id": "user1"
10    },
11    {
12      "amount": 1,
13      "message": "msg1",
14      "timestamp": 1776631115.948956,
15      "transaction_id": "msg1",
16      "user_id": "user1"
17    },
18    {
19      "amount": 1,
20      "message": "msg5",
21      "timestamp": 1776631153.215095,
22      "transaction_id": "msg5",
23      "user_id": "user1"
24    },
25    {
26      "amount": 1,
27      "message": "msg3",
28      "timestamp": 1776631143.5400674,
29      "transaction_id": "msg3",
30      "user_id": "user1"
31    },
32    {
33      "amount": 1,
34      "message": "msg6",
35      "timestamp": 1776631159.0957696,
36      "transaction_id": "msg6",
37      "user_id": "user1"
38    },
39    {
40      "amount": 1,
41      "message": "msg9",
42      "timestamp": 1776631175.1699586,
43      "transaction_id": "msg9",
44      "user_id": "user1"
45    }
46  ]
47 }
```

Бачимо що транзакції виконано, тепер перевіримо логи кожного logging service:

```

akravchuk@macA apz-kravchuk % docker logs apz-kravchuk-logging-service-1-1
[2026-04-19 20:37:29 +0000] [1] [INFO] Starting gunicorn 22.0.0
[2026-04-19 20:37:29 +0000] [1] [INFO] Listening at: http://0.0.0.0:8001 (1)
[2026-04-19 20:37:29 +0000] [1] [INFO] Using worker: sync
[2026-04-19 20:37:29 +0000] [7] [INFO] Booting worker with pid: 7
[2026-04-19 20:37:29 +0000] [8] [INFO] Booting worker with pid: 8
[logging-service:1] stored tx=msg4 user=user1 amount=1 msg=msg4
[logging-service:1] stored tx=msg5 user=user1 amount=1 msg=msg5
[logging-service:1] stored tx=msg7 user=user1 amount=1 msg=msg7
[logging-service:1] stored tx=msg9 user=user1 amount=1 msg=msg9
[logging-service:1] stored tx=msg10 user=user1 amount=1 msg=msg10
akravchuk@macA apz-kravchuk % docker logs apz-kravchuk-logging-service-2-1
[2026-04-19 20:37:29 +0000] [1] [INFO] Starting gunicorn 22.0.0
[2026-04-19 20:37:29 +0000] [1] [INFO] Listening at: http://0.0.0.0:8001 (1)
[2026-04-19 20:37:29 +0000] [1] [INFO] Using worker: sync
[2026-04-19 20:37:29 +0000] [7] [INFO] Booting worker with pid: 7
[2026-04-19 20:37:29 +0000] [8] [INFO] Booting worker with pid: 8
[logging-service:2] stored tx=msg1 user=user1 amount=1 msg=msg1
[logging-service:2] stored tx=msg3 user=user1 amount=1 msg=msg3
[logging-service:2] stored tx=msg6 user=user1 amount=1 msg=msg6
akravchuk@macA apz-kravchuk % docker logs apz-kravchuk-logging-service-3-1
[2026-04-19 20:37:29 +0000] [1] [INFO] Starting gunicorn 22.0.0
[2026-04-19 20:37:29 +0000] [1] [INFO] Listening at: http://0.0.0.0:8001 (1)
[2026-04-19 20:37:29 +0000] [1] [INFO] Using worker: sync
[2026-04-19 20:37:29 +0000] [7] [INFO] Booting worker with pid: 7
[2026-04-19 20:37:29 +0000] [8] [INFO] Booting worker with pid: 8
[logging-service:3] stored tx=msg2 user=user1 amount=1 msg=msg2
[logging-service:3] stored tx=msg8 user=user1 amount=1 msg=msg8
akravchuk@macA apz-kravchuk %

```

Бачимо, що сервіс 1 опрацював транзакції 4, 5, 7, 9 та 10, сервіс 2 опрацював 1, 3 і 6, а сервіс 3 опрацював 2 і 8. Так як кожен меседж був з amount = 1, то баланс має бути 10, що показує скрін вище.

Тепер перевіримо відмовостійкість і виключимо counter-service:

```

akravchuk@macA apz-kravchuk % docker pause apz-kravchuk-counter-service-1
apz-kravchuk-counter-service-1
akravchuk@macA apz-kravchuk % docker ps
CONTAINER ID   IMAGE                                COMMAND                  CREATED        STATUS        PORTS                               NAMES
6605513963a1  apz-kravchuk-facade-service         "gunicorn -w 4 -b 0..." 7 minutes ago  Up 7 minutes  0.0.0.0:8000->8000/tcp             apz-kravchuk-facade-service-1
4c52320e48ea  apz-kravchuk-logging-service-3      "gunicorn -w 2 -b 0..." 7 minutes ago  Up 7 minutes  0.0.0.0:8004->8001/tcp             apz-kravchuk-logging-service-3-1
201f82badf8a  apz-kravchuk-logging-service-1      "gunicorn -w 2 -b 0..." 7 minutes ago  Up 7 minutes  0.0.0.0:8001->8001/tcp             apz-kravchuk-logging-service-1-1
f187e61dc244  apz-kravchuk-logging-service-2      "gunicorn -w 2 -b 0..." 7 minutes ago  Up 7 minutes  0.0.0.0:8003->8001/tcp             apz-kravchuk-logging-service-2-1
e5fa7481dc37  apz-kravchuk-counter-service        "gunicorn -w 2 -b 0..." 7 minutes ago  Up 7 minutes  0.0.0.0:8002->8002/tcp             apz-kravchuk-counter-service-1
421909e204f5  hazelcast/hazelcast:5.3            "hz start"              7 minutes ago  Up 7 minutes  5701/tcp                           apz-kravchuk-hazelcast1-1
ae25151d5373  apache/kafka:3.9.0                 "/_cacert Entrypoin..." 7 minutes ago  Up 7 minutes  0.0.0.0:9092->9092/tcp             kafka
ea26ad7606ff  postgres:16-alpine                 "docker-entrypoint.s..." 7 minutes ago  Up 7 minutes  5432/tcp                           apz-kravchuk-postgres-1
33a91a993fa0  hazelcast/hazelcast:5.3            "hz start"              7 minutes ago  Up 7 minutes  5701/tcp                           apz-kravchuk-hazelcast2-1
51d0867d8e31  hazelcast/hazelcast:5.3            "hz start"              7 minutes ago  Up 7 minutes  5701/tcp                           apz-kravchuk-hazelcast3-1
75673642bcdb  apz-kravchuk-config-server         "gunicorn -w 2 -b 0..." 7 minutes ago  Up 7 minutes  0.0.0.0:8010->8010/tcp             apz-kravchuk-config-server-1
akravchuk@macA apz-kravchuk %

```

Бачимо що counter зупинений, тепер пробуємо відправити меседж:

HTTP <http://localhost:8000/transaction>

POST [▼](#) <http://localhost:8000/transaction>

[Docs](#) [Params](#) [Authorization](#) [Headers \(9\)](#) **[Body](#)** [Scripts](#) [Settings](#)

☐ none ☐ form-data ☐ x-www-form-urlencoded ☒ raw ☐ binary ☐ GraphQL **JSON**

```
1  {"user_id":"user1","amount":1,"transaction_id":"msg11","message":"msg11"}
2
```

Body [Cookies \(1\)](#) [Headers \(5\)](#) [Test Results](#)

{ } JSON [▼](#) [▶ Preview](#) [🖼 Visualize](#) [▼](#)

```
1  {
2    "queued": true,
3    "transaction_id": "msg11"
4  }
```

Бачимо що транзакція успішна, тепер перевіримо GET запит:

HTTP <http://localhost:8000/user/user1>

GET <http://localhost:8000/user/user1>

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Query Params

Key	Value
Key	Value

Body Cookies (1) Headers (5) Test Results

{ } JSON Preview Debug with AI

```
1
2 "details": "HTTPConnectionPool(host='counter-service', port=8002): Read timed out. (read timeout=10)",
3 "error": "counter-service unreachable"
4
```

Тепер повертаємо сервіс до ладу і знову пробуємо GET запит:

```
akravchuk@macA apz-kravchuk % docker unpause apz-kravchuk-counter-service-1
apz-kravchuk-counter-service-1
akravchuk@macA apz-kravchuk % docker ps
CONTAINER ID   IMAGE                                COMMAND                  CREATED        STATUS        PORTS                               NAMES
6605513963a1  apz-kravchuk-facade-service         "gunicorn -w 4 -b 0..." 12 minutes ago Up 12 minutes 0.0.0.0:8000->8000/tcp             apz-kravchuk-facade-service-1
4c52320e48ea  apz-kravchuk-logging-service-3      "gunicorn -w 2 -b 0..." 12 minutes ago Up 12 minutes 0.0.0.0:8004->8001/tcp             apz-kravchuk-logging-service-3-1
201f82badf8a  apz-kravchuk-logging-service-1      "gunicorn -w 2 -b 0..." 12 minutes ago Up 12 minutes 0.0.0.0:8001->8001/tcp             apz-kravchuk-logging-service-1-1
f8f7e61dc244  apz-kravchuk-logging-service-2      "gunicorn -w 2 -b 0..." 12 minutes ago Up 12 minutes 0.0.0.0:8003->8001/tcp             apz-kravchuk-logging-service-2-1
e5fa7481dc37  apz-kravchuk-counter-service        "gunicorn -w 2 -b 0..." 12 minutes ago Up 12 minutes 0.0.0.0:8002->8002/tcp             apz-kravchuk-counter-service-1
421909e204f5  hazelcast/hazelcast:5.3            "hz start"              12 minutes ago Up 12 minutes 5701/tcp                          apz-kravchuk-hazelcast1-1
ae25151d5373  apache/kafka:3.9.0                 "/_cacert_entrypoint..." 12 minutes ago Up 12 minutes 0.0.0.0:9092->9092/tcp             kafka
ea26ad7606ff  postgres:16-alpine                 "docker-entrypoint.s..." 12 minutes ago Up 12 minutes 5432/tcp                          apz-kravchuk-postgres-1
33a91a993ffa  hazelcast/hazelcast:5.3            "hz start"              12 minutes ago Up 12 minutes 5701/tcp                          apz-kravchuk-hazelcast2-1
51d0867d8e31  hazelcast/hazelcast:5.3            "hz start"              12 minutes ago Up 12 minutes 5701/tcp                          apz-kravchuk-hazelcast3-1
75673642bcdb  apz-kravchuk-config-server         "gunicorn -w 2 -b 0..." 12 minutes ago Up 12 minutes 0.0.0.0:8010->8010/tcp             apz-kravchuk-config-server-1
akravchuk@macA apz-kravchuk %
```

Бачимо що counter-service знову працює, тому рухаємося далі:

HTTP <http://localhost:8000/user/user1>

GET

<http://localhost:8000/user/user1>

Docs Params Authorization Headers (7) Body Scripts Settings

Body Cookies (1) Headers (5) Test Results

{ } JSON

Preview

Visualize

```
1  {
2    "balance": 11,
3    "transactions": [
4      {
5        "amount": 1,
6        "message": "msg2",
7        "timestamp": 1776631138.7859244,
8        "transaction_id": "msg2",
9        "user_id": "user1"
10     },
11     {
12       "amount": 1,
13       "message": "msg1",
14       "timestamp": 1776631115.948956,
15       "transaction_id": "msg1",
16       "user_id": "user1"
17     },
18     {
19       "amount": 1,
20       "message": "msg5",
21       "timestamp": 1776631153.215095,
22       "transaction_id": "msg5",
23       "user_id": "user1"
24     },
25     {
26       "amount": 1,
27       "message": "msg3",
28       "timestamp": 1776631143.5400674,
29       "transaction_id": "msg3",
30       "user_id": "user1"
31     },
32     {
33       "amount": 1,
34       "message": "msg6",
35       "timestamp": 1776631159.0957696,
36       "transaction_id": "msg6",
37       "user_id": "user1"
38     },
39     {
40       "amount": 1,
41       "message": "msg9",
42       "timestamp": 1776631175.1699586,
43       "transaction_id": "msg9",
44       "user_id": "user1"
45     }
46   ]
47 }
```

HTTP <http://localhost:8000/user/user1>

GET <http://localhost:8000/user/user1>

Docs Params Authorization Headers (7) Body Scripts Settings

Body Cookies (1) Headers (5) Test Results

{ } JSON Preview Visualize

```
39 {
40   "amount": 1,
41   "message": "msg9",
42   "timestamp": 1776631175.1699586,
43   "transaction_id": "msg9",
44   "user_id": "user1"
45 },
46 {
47   "amount": 1,
48   "message": "msg10",
49   "timestamp": 1776631180.737052,
50   "transaction_id": "msg10",
51   "user_id": "user1"
52 },
53 {
54   "amount": 1,
55   "message": "msg8",
56   "timestamp": 1776631169.4903138,
57   "transaction_id": "msg8",
58   "user_id": "user1"
59 },
60 {
61   "amount": 1,
62   "message": "msg4",
63   "timestamp": 1776631148.0558822,
64   "transaction_id": "msg4",
65   "user_id": "user1"
66 },
67 {
68   "amount": 1,
69   "message": "msg7",
70   "timestamp": 1776631164.6893582,
71   "transaction_id": "msg7",
72   "user_id": "user1"
73 },
74 {
75   "amount": 1,
76   "message": "msg11",
77   "timestamp": 1776631544.2579257,
78   "transaction_id": "msg11",
79   "user_id": "user1"
80 }
81 ]
82 }
```


Бачимо що баланс оновився і в кінці в списку транзакцій є msg11, отже все працює добре.